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Nonlinear Aeroelastic Analysis of High-Aspect-Ratio Wings with a Low-Order Propeller Model

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A nonlinear aeroelastic analysis framework for high-aspect-ratio wings that includes the aerodynamic effects of propellers is described. The high computational cost required for modeling aerodynamic interaction between the wing and propeller wake is reduced by taking advantage of the relatively slow dynamics of the wing. Consequently, the propeller wake is modeled as a straight vortex cylinder that does not require a computationally expensive wake updating process. By leveraging the smallness of the propeller, an averaged vortex cylinder method is proposed that calculates the induced velocities of the propeller vortex cylinder efficiently without suffering from a numerical singularity and loss of accuracy. The induced velocities are considered in the wing aerodynamic force calculation using an unsteady vortex lattice method. An efficient propeller cylinder coordinate generation method modeling the propeller-attached wing by absolute nodal coordinate formulation with a multibody dynamic theory is also proposed. The developed framework is validated by comparison with other formulations. A static aeroelastic analysis on a low-speed high-aspect-ratio wing demonstrates that the propeller-induced axial velocity causes the deflection change of 5.4%. A nonlinear dynamic result shows that the propeller decreases the vibration amplitude by 6.7% because the propeller-induced axial velocity enhances the aerodynamic damping.

Nomenclature

$\boldsymbol{A}$	=	aerodynamic influence coefficient matrix
a'	=	tangential induction factor
C_l, C_d, C_T	=	lift, drag, and propeller thrust coefficients
$\boldsymbol{e}$	=	generalized nodal coordinate vector
$\boldsymbol{e}_a$, $\boldsymbol{e}_t$	=	base vectors of propeller cylinder coordinate system
$\boldsymbol{F}$	=	generalized external force vector
$\boldsymbol{F}_{\text{aero}}$	=	generalized aerodynamic force vector
$\boldsymbol{F}_{\text{elastic}}$	=	generalized elastic force vector
$\boldsymbol{K}$	=	linearized stiffness matrix
$\boldsymbol{M}$	=	mass matrix
$\boldsymbol{n}$	=	vector normal to aerodynamic panel
R	=	propeller radius
$\boldsymbol{r}$	=	position vector of an arbitrary point
$\boldsymbol{r}_x$, $\boldsymbol{r}_y$, $\boldsymbol{r}_z$	=	gradient vectors
T	=	thrust
U	=	flow velocity normal to propeller plane
$\boldsymbol{U}$	=	uniform flow velocity
u_a, u_t	=	propeller-induced axial and tangential velocities
$\boldsymbol{w}$	=	local velocity due to wake panel, propeller, dynamic motion, and freestream
α	=	pitch angle of propeller or angle of attack
β	=	yaw angle of propeller
Γ_p	=	circulation distribution along propeller blade
Γ_{tot}	=	circulation of propeller vortex cylinder
$\boldsymbol{\gamma}$	=	terms obtained from the second-order differential of the constraint equation vector
γ_t	=	circulation of propeller tangential vortex
δ	=	cutoff radius coefficient
$\boldsymbol{\lambda}$	=	Lagrange multiplier
ρ_a	=	air density
ψ	=	azimuthal angle of propeller cylinder coordinate system
$\boldsymbol{\Phi}$	=	constraint equation vector
$()_{\text{ECE}}$	=	value related to external constraint equation
$()_{\text{ICE}}$	=	value related to internal constraint equation
$()_L$	=	value defined in local coordinate system
$()^i$	=	value related to node i (is equal to A, B, P, Q)

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I. Introduction

ELECTRIC aircraft with distributed propellers are now considered for a variety of applications. Common features include very-high-aspect-ratio wings and propeller-based thrust generation [1], as shown in Fig. 1. The Helios Prototypes [2] are representative early examples of vehicle in this class. Helios Prototypes HP01 and HP03 had 14 and 10 propellers, respectively. Although each propeller is relatively small, the wake of many propellers covered a large area of the main wing, resulting in an aerodynamic propeller–wing interaction. High-aspect-ratio wings display high flexibility [3] that cannot be expressed by using linear analysis methods, as shown in Fig. 1. Therefore, the nonlinear static and dynamic characteristics also need to be considered in their design.

The study of the aerodynamic propeller–wing interaction, without considering wing deformations, has a long history. In conventional studies, vortex lattices or panel methods have been widely used for the aerodynamic modeling of both the propeller and wing since the 1980s. Rangwalla and Wilson [4] presented the circulation distribution on the propeller blades and wing obtained from the aerodynamic propeller–wing analysis using the vortex lattice method. Witkowski et al. [5] also used the vortex lattice method for the analysis and showed that both axial and tangential velocities induced by the propeller have a significant effect on the wing lift distribution. The axial induced velocity increases the wing lift. The tangential induced velocity increases and decreases the wing lift on either side of the propeller. Later studies [6,7] compared the analysis using the vortex

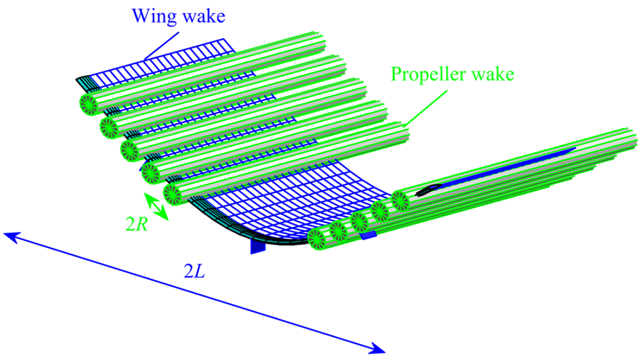


Fig. 1 Conceptual diagram of high-aspect-ratio-wing aircraft with propellers.

lattice method or a higher-order vortex method with experimental results. They showed that lift coefficients were in good agreement. More recently, a viscous vortex particle method (VVPM) has been investigated for the propeller wake modeling [8]. This is because the VVPM can avoid the singularity problem of the vortex lattice method when propeller wake panels approach to wing panels.

All the studies above were for rigid aircraft. Teixeira and Cesnik [1,9,10] have recently pioneered the integration of propeller aerodynamics in nonlinear aeroelasticity. In their framework, the aerodynamics of propeller and propeller wake are modeled by the lifting line theory and the VVPM (LL/VVPM), respectively. The wing aerodynamics are modeled by unsteady vortex lattice method (UVLM). Not only the propeller aerodynamics but also gyroscopic effects were considered. They analyzed the nonlinear static and dynamic responses of high-aspect-ratio-wing aircraft. Their static analysis indicated that the deformation of the main wing increases due to the propeller effect. Their dynamic analysis indicated that the propeller gyroscopic effect is negligibly small on typical lightweight high-aspect-ratio-wing aircraft. However, the computational cost becomes high as the propeller rotation speed is much higher than the flight speed. As a result, there is a large number of propeller wake elements close to the wing that have nonnegligible effect on the wing aerodynamics. Additionally, the time step of the propeller wake shedding needs to be small to deal with the high-speed rotation and to enforce the nonpenetration boundary condition. Here, the propeller of the high-aspect-ratio-wing aircraft is often operated at a high rotational speed [1] although the aircraft flies at a low speed, and this will create a computational bottleneck when many parametric analyses are necessary. In summary, the gap that prevented us from considering the propeller in an aeroelastic analysis of the high-aspect-ratio-wing aircraft is a computational cost. The low-speed flight and the high-speed propeller rotation require many wake elements (i.e., vortex particle or vortex panel). The high computational cost caused by the many wake elements prevented us from considering the propeller effects in the aeroelastic analysis especially at an early design stage. To consider the propeller effect in the aeroelastic analysis at an earlier design stage, the gap is worth filling.

Alba et al. [11] and An [12] modeled the propeller aerodynamic effects on rigid and flexible wings by using the actuator disk theory with a vortex cylinder model that consists of tangential, longitudinal, root, and bound vortex components [13]. Although this modeling method is highly efficient because it does not require the wake shedding, it considers the propeller as an ideal actuator disk neglecting the propeller geometry information (e.g., number of propeller blades and blade shape) and it assumes an elliptic radial circulation distribution on the actuator disk. More recently, Branlard and Gaunaa [14] have presented another computationally efficient vortex cylinder method (VCM) assuming a constant radial circulation distribution. Once the propeller circulation is provided, the propeller-induced velocities at an arbitrary point are calculated analytically. This method can be extended to consider the effects of the skewed flow [15] and unsteadiness [16]. Additionally, the constant circulation assumption can be relaxed by superposing co-axial vortex cylinders if necessary.

For structural modeling of the high-aspect-ratio wing, nonlinear beam formulations have been used [17] such as a displacement-based geometrically exact beam formulation [18,19], a strain-based beam formulation [2], an intrinsic beam formulation [20], and a corotational beam formulation [21,22]. A rigid-multibody formulation with spring joints [23] was also used. In this paper, we focus on another formulation, namely, absolute nodal coordinate formulation (ANCF) [24] to develop a nonlinear static and dynamic analysis framework considering propellers. In ANCF, position and gradient vectors are used as variables to describe large rigid-body motion and very flexible deformation. ANCF provides various finite elements including not only beam [25] but also plate [26] and solid [27] with a constant mass matrix and linear external constraint equations (ECEs) to express mechanical joints used for wings, such as fixed and hinge joints. However, the norm and inner product variation of the gradient vectors of the ANCF beam element used for aerospace applications [28] compulsorily express cross-sectional deformations that cause numerical locking problems [29]. Otsuka and Makihara [30] have developed an ANCF with internal constraint equations (ANCF-ICE) that constrain the problematic cross-sectional deformations (i.e., retain orthogonality and unity of the gradient vectors) by a multibody dynamic theory. The nonlinear aeroelastic dynamic analysis using the ANCF-ICE structural model coupled with an UVLM aerodynamic model has been validated by wind tunnel experiments [31].

The objective of this paper is to develop an efficient nonlinear static and dynamic analysis framework of the high-aspect-ratio wing with propellers. To achieve this objective, we solve the problem of the aerodynamic propeller–wing interaction by leveraging the characteristics of the high-aspect-ratio wing with propellers. The first characteristic is that dominant frequencies of the wing are relatively low. We focus on modeling the quasi-steady physics occurring on the aircraft of our interest. We assume that the propeller wake is a cylinder always aligned with the propeller in both the static and dynamic analyses, as shown in Fig. 1. This assumption may not be valid in the cases where the wake is deformed largely or quickly by close propellers, sideslip, gust, and torsion. However, Branlard and Gaunna [15,16] extended VCM to skewed flow conditions recently. Therefore, our approach based on VCM has a potential to be used even when quick sideslip, strong gust, and large torsion deform the wake largely or quickly. This extension potentially enables the proposed method to be applied to multiple lifting surfaces behind the main wing even when the twisted main wing deviates the cylinder wake. This vortex cylinder model enables us to use a computationally efficient calculation method of the propeller-induced velocities because it does not require the propeller wake updating process. To calculate the propeller-induced velocities more efficiently, we propose an averaged vortex cylinder method (AVCM) by leveraging a second characteristic, namely, a small propeller–wing ratio that we define as the propeller radius R divided by the wing span L (e.g., $R/L = 0.027$ for Helios Prototype HP03). Because of the small R/L, the radial variation of the propeller circulation can be negligible when considering their effect on the wing. Therefore, the propeller circulation is considered as a constant and averaged value. To calculate the propeller circulation, we employ a computationally efficient blade element momentum (BEM) theory that incorporates the propeller geometry information that would be neglected in the conventional propeller–wing interaction study using the VCM. The proposed BEM/AVCM provides the propeller-induced velocities that affect the wing aerodynamics modeled by the UVLM. By leveraging the orthogonal and unit gradient vectors of the ANCF-ICE obtained by using a multibody dynamic theory, we also propose a method to efficiently generate the propeller vortex cylinder coordinate system even when the propeller position largely changes owing to the nonlinearly large deformation.

The proposed framework is compared with other formulations to verify its accuracy and effectiveness. Then, we investigate the propeller effects on the nonlinear static and dynamic responses of a high-aspect-ratio wing. We solve a similar problem to Teixeira and Cesnik [1]. However, we propose a very different solution process tailored to problems with slow dynamics, which are much common on the platforms we investigate. As a result, the propeller aerodynamics, propeller wake, and structural modeling are completely different and the main similarity is the UVLM used for the wing aerodynamics.

Our work has three novelties obtained from the differences as follows:

First, we employed VCM without the wake discretization by focusing on the slow dynamics of the high-aspect-ratio wings, whereas Teixeira and Cesnik [1] employed VVPM with the wake discretization. The aircraft of our interest have low flight speed and high propeller rotation speed, which results in a nonnegligible propeller wake on the main wing. A VVPM discretization, however, results in a high computational cost regardless even for steady flight. This first novelty significantly reduces the calculation time. Several research groups [32,33] have developed similar approaches coupling a propeller actuator disk model with a high-fidelity computational fluid dynamics. Although the unsteadiness caused by the propeller rotation was considered [33], the aeroelasticity and the associated unsteady propeller movement (e.g., a heaving motion of a propeller center) were not taken into account.

The second novelty is the proposal of AVCM leveraging the small propeller–wing ratio R/L. AVCM enables a faster calculation by eliminating the vortex superposition used in a conventional VCM. This second novelty reduces the calculation time furthermore.

The third novelty is the proposal of the efficient propeller cylinder coordinate generation method leveraging ANCF-ICE. ANCF-ICE has orthonormal vectors as nodal variables. The orthonormal vectors can be directly used to obtain the propeller cylinder coordinate system.

Clearly, our approach is only valid for sufficiently slow dynamics, beyond which a VVPM or similar wake updating approach would be needed. Here, we define the approach applying the vortex lattice or panel method to both the wing and propeller as a conventional vortex-based method. Numerous studies [4–6] have employed the vortex-based method to investigate the propeller–wing interaction effects on rigid-wing aerodynamics. Although the conventional vortex-based method is straightforward, it still requires many wake elements, resulting in a high computational cost. Moreover, the conventional vortex-based method is more likely to suffer from a singularity than our low-cost approach and the high-cost VVPM [1].

II. Aerodynamic Propeller–Wing Interaction Model

The propeller–wing interaction modeling in the proposed framework is presented in this section. First, the propeller aerodynamics is modeled by the proposed AVCM coupled with the BEM theory. Then, the UVLM for the wing aerodynamics is briefly summarized. Finally, the propeller–wing interaction modeling is presented. To obtain a better computational efficiency, the proposed method is based on several assumptions.

The assumptions used for the propeller aerodynamics are as follows:

1) Propeller wake is modeled as a vortex cylinder. This assumption may not hold in some cases where the wake deforms largely (e.g., close propellers, strong gust, quick side slip, and large torsion).

2) Skewed flow effect is neglected.

3) The circulation of a propeller blade is a constant averaged value.

4) Propeller-induced axial and tangential flows of a propeller are time-averaged values.

5) A propeller-induced tangential flow of a propeller is a space-averaged value.

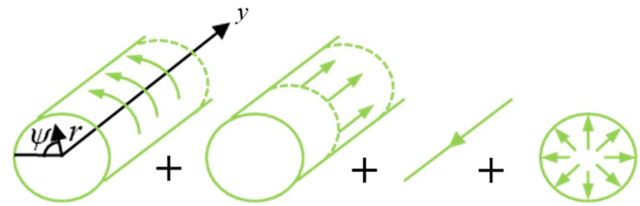


Fig. 2 Vortex cylinder model and cylinder coordinate system for propeller aerodynamics.

6) A propeller-induced flow behind a hub is zero.

The assumptions used for the wing aerodynamics are as follows:

1) The wing aerodynamics is based on UVLM with a potential flow assumption.

2) The roll up of the wing wake is neglected.

The assumption used for the propeller–wing aerodynamic interaction are as follows:

1) One-way coupling assumption: The propeller-induced flow has an effect on the wing aerodynamics, whereas the wing-induced flow does not have an effect on the propeller aerodynamics.

2) Propeller-induced velocities far from the vortex cylinder are cut off.

A. Blade Element Momentum Theory and Averaged Vortex Cylinder Method for Propeller Aerodynamics

The propeller-induced velocities that affect the wing aerodynamics are calculated by combining BEM and AVCM. In the proposed AVCM, the propeller wake is modeled as a vortex cylinder [14]. A vortex cylinder consists of tangential, longitudinal, root, and bound vortex components, as shown in Fig. 2. This modeling provides the propeller-induced velocities analytically by assuming no propeller wake deformation. The length scale of the wake in the streamwise direction is longer than that of the distance between the wing and propeller. Therefore, the wake deformation has a small effect on the wing aerodynamics even when the wake is deformed by the sideslip or gust. As it was discussed in Sec. I, however, considering the skewed flow effect is necessary when quick sideslip and strong gusts occur. Moreover, the presence of the wing varies the propeller tangential velocity [34]. This tangential velocity variation is not taken into account by the proposed method because we assume that the propeller–wing ratio R/L is small. The tangential velocity variation would have a larger effect on the aeroelastic responses of the wing if the propeller–wing ratio R/L is large. This is because the tangential velocity causing upwash and downwash changes the internal moment of the wing largely when R/L is large. Investigating the relationship between R/L and the tangential velocity varied by the presence of the wing is one of the future studies. Another assumption of this approach is the constant circulation on the propeller plane, which is inspired by the small propeller–wing ratio R/L. Although the axial and tangential velocities induced by the vortex cylinder can be calculated analytically [16], the analytical induced velocities have a singularity at a propeller center $r = 0$. We assume that the axial velocity behind a propeller hub with a radius R_c are zero to avoid the singularity. We also consider the tangential velocity as an averaged value evaluated at $r = R/2$. By using these assumptions and the Biot–Savart law, the induced velocities of the vortex cylinder in the proposed AVCM can be obtained as

$$
u_a \equiv \begin{cases} 0 & ;\ r \le R_c \\ \dfrac{\gamma_t}{2}\left[\dfrac{R - r + |R - r|}{2|R - r|} + \dfrac{y k(y)}{2\pi\sqrt{rR}}\left(K(k^2(y)) + \dfrac{R - r}{R + r}\Pi(k^2(0), k^2(y))\right)\right] & ;\ R_c < r \end{cases}
$$

$$
u_t \equiv \begin{cases} -\dfrac{\Gamma_{\text{tot}}}{2\pi(R/2)} & ;\ R_c \le r \\ 0 & ;\ r > R \end{cases} \tag{1}
$$

where

$$k^2(y) \equiv \frac{4rR}{(R+r)^2 + y^2} \tag{2}$$

r and y are radial and axial coordinates in the cylinder coordinate system. K and Π are elliptic integrals of the first and third kinds, respectively. The Biot–Savart law defined in the cylindrical coordinate system includes a square root of trigonometric functions, and thus the elliptic integrals appear, as shown in [14]. Γ_{tot} and γ_t are tangential and longitudinal circulations. The direction of the propeller rotation is taken into account by considering the positive/negative sign of the blade circulation in Eq. (1). To demonstrate that the proposed AVCM can avoid the singularity unlike the original VCM, the propeller-induced velocities normalized by the tangential and longitudinal circulations are compared with the reference values calculated by the original VCM [14]. Figure 3 shows the propeller-induced axial and tangential velocities R and $0.1R$ behind the propeller, respectively. Apart from the propeller hub region, good agreement of the axial velocity is found in Fig. 3a, which also does not have singularities. This is because the singularity avoidance method presented by Branlard and Gaunna [14] was employed for Eq. (1). The first term of Eq. (1) has a singularity at $r/R = 1$. Therefore, the first term is considered as a mean value of the induced flow velocities in the outer and inner parts. The mean value is $1/2$. The singularity of the tangential velocity in Fig. 3b can be avoided by the proposed BEM/ACVM. Averaging the tangential velocity is reasonable because the radial variation effect of the tangential velocity on the wing aerodynamics is small because of the small propeller–wing ratio R/L. The validity of averaging the tangential velocity in the propeller–wing interaction analysis will be shown in Sec. V.

To obtain the nonnormalized propeller-induced velocities, Γ_{tot} and γ_t need to be calculated. In the proposed AVCM, the constant tangential circulation Γ_{tot} is defined as a radially averaged value:

$$\Gamma_{\text{tot}} \equiv \frac{\int_0^R \Gamma_p(r)\,\mathrm{d}r}{R} \approx \frac{\sum_{i=1}^{N} (\Gamma_p)_i}{N} \tag{3}$$

This equation includes a discrete form to combine AVCM and BEM that discretizes the propeller blade in N equal segments. The index i represents the ith propeller blade segments. The tangential circulation distribution can be related to the tangential induction factor as

$$(\Gamma_p)_i = 4\pi\Omega r^2 a_i' \tag{4}$$

where Ω is a propeller rotation speed. In BEM, the induction factor is calculated efficiently in an iteration process by using the propeller blade data sets. The data sets are the twist and chord distributions along the blade and the aerodynamic coefficients corresponding to the airfoil shape. Although the BEM method has been extended to consider the various unsteady effects, only unsteady variation of the input velocity is considered in first approximation. Once the tangential circulation is obtained, the longitudinal circulation is calculated as [16]

$$\gamma_t = \sqrt{U^2 - \frac{\Gamma_{\text{tot}}^2}{4\pi^2 R^2} - \frac{\Omega \Gamma_{\text{tot}}}{\pi}} - U \tag{5}$$

Here, we summarize the limitations of the proposed BEM/AVCM as follows:

First, the accuracy of the proposed BEM/AVCM may decrease as the distance between a propeller and a lifting surface (e.g., tail wing) increases, although the proposed BEM/AVCM can be applied to not only isolated wings but also multiple lifting surfaces. This is because the propeller wake deforms as it travels in a long distance. It is important to carefully investigate the validity of neglecting the propeller wake deformation as a future study before the proposed method is applied to multiple lifting surfaces.

Second, our current implementation neglects the skewed flow effect and thus we assume a flight without a quick sideslip or strong gust (i.e., skewed flow effects). We note, however, that the VCM was extended to consider the skewed flow effects by Branlard and Gaunna [15], which we intend to explore in the near future.

Third, a limitation is caused by BEM used for the proposed method. BEM requires sectional aerodynamic coefficients obtained from experimental, analytical, or computational methods.

B. Unsteady Vortex Lattice Method for Wing Aerodynamics

Wing aerodynamics are modeled by using UVLM. In UVLM, each lifting surface is discretized into M bound vortex panels, as shown in Fig. 4. The bound panel $j(j = 1, \ldots, M)$ has a circulation Γ_j. Wake panels are shed from the trailing edge at each time step. The circulation of the wake panels is convected from the trailing edge panels enforcing the Kutta condition. Therefore, only the bound vortex circulations are treated as unknown variables. The bound vortex circulations are calculated by using the nonpenetration relationship that means that the relative flow velocity normal to a bound vortex panel is zero at the collocation points located at the quarter chordwise dimension of each panel. The nonpenetration relationship can be written as

$$\begin{bmatrix} \Gamma_1 \\ \Gamma_2 \\ \vdots \\ \Gamma_M \end{bmatrix} = \boldsymbol{A}^{-1} \begin{bmatrix} \boldsymbol{w}_1^T \boldsymbol{n}_1 \\ \boldsymbol{w}_2^T \boldsymbol{n}_2 \\ \vdots \\ \boldsymbol{w}_M^T \boldsymbol{n}_M \end{bmatrix} \tag{6}$$

The aerodynamic coefficient matrix $\boldsymbol{A}$ is obtained from Biot–Savart law; $\boldsymbol{w}_j$ is a local flow velocity vector including the freestream velocity,

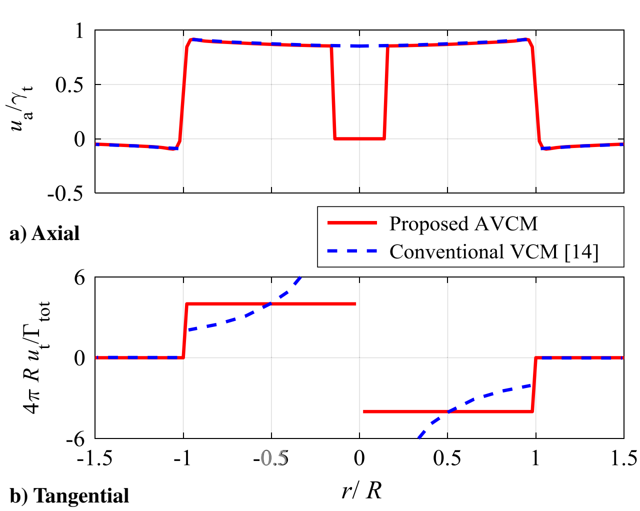


Fig. 3 a) Axial and b) tangential induced velocities calculated by proposed AVCM and conventional VCM.

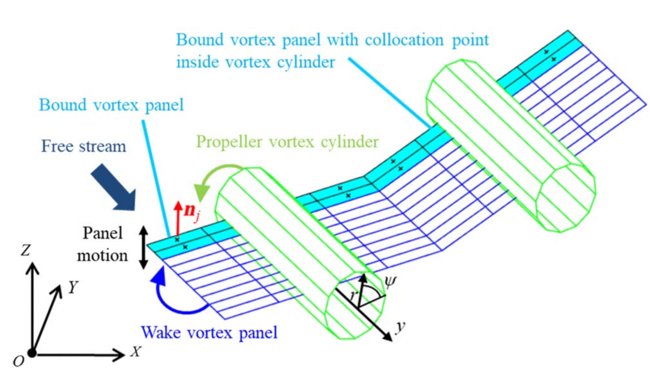


Fig. 4 Propeller–wing aerodynamic interaction modeling.

wake-induced velocity, panel motion velocity, and propeller-induced velocity calculated by Eq. (1). An overestimation of the propeller-induced velocities may occur at a bound vortex panel when its collocation point is located inside a vortex cylinder, as shown in Fig. 4. This is because the induced velocities drastically change at $r/R = 1$ and $r/R = -1$, as shown in Fig. 3. To mitigate the overestimation, we employ two approaches. First, the spanwise bound vortex discretization behind the propeller plane is relatively fine. Second, propeller-induced velocities are evaluated at four corner points of a vortex ring, and then the mean value of the velocities is defined as the propeller-induced velocity at a collocation point. Once the bound vortex circulation is determined, the aerodynamic force on the bound vortex panel is calculated by the Kutta–Joukowski method [35]. The wing wake panels are updated at each time step based on the local flow velocity including $\boldsymbol{w}$ plus induced velocity of the bound vortex panels.

C. Propeller–Wing Interactive Aerodynamics

Potentially high computational costs of the propeller–wing aerodynamic interaction analysis become a concern especially when the structural deformation is considered. Therefore, two assumptions are introduced in the propeller–wing interaction to reduce the computational cost.

First, we assume that the induced velocity of the wing does not affect the propeller aerodynamics (i.e., one-way coupling [12]). The limitation caused by the one-way coupling is neglecting the wing-induced flow effect on the propeller circulation. Marretta [36] investigated the effect of the wing on the propeller performance. Because the wing circulation causes an induced flow even if the propeller is located in front of the propeller, the propeller thrust (i.e., propeller circulation) varies. However, we can assume that the wing-induced velocities on the propeller is negligible when the propeller rotation speed (i.e., propeller circulation caused by the propeller rotation) is higher than the flight speed (i.e., propeller circulation caused by the wing-induced flow). Section V will show that the proposed method with the one-way coupling assumption is in good agreement with a method that does not have the assumption. If necessary, this limitation can be addressed by considering the wing-induced flow calculated by the UVLM in the propeller aerodynamic solver.

Second, we assume that the induced velocity of the propeller does not affect the bound and wake panels far from the propeller vortex cylinder. Therefore, the propeller-induced velocity is neglected if r is more than a cutoff distance $\delta \times R$. We determined the cutoff distance as $2R$ in this paper. This is because the propeller-induced velocities outside of the propeller radius are several orders less than those inside of the propeller radius, as shown in Fig. 3.

In this paper, the load transfer from the propeller to the wing is not taken into account because the focus of this paper is the aerodynamic propeller–wing interaction effect on the high-aspect-ratio-wing aeroelasticity. However, the load transfer may affect the wing deformation and local angle of attack, and it is worth investigating it as a future study. The consideration of the load transfer is possible by using the orthonormal gradient vectors, as shown in the author's previous work [37].

III. Flexible Wing Model

This section presents the structural modeling based on the ANCF with internal constraint equations (ANCF-ICE) [30] that solves the numerical locking problems by retaining the orthogonality and unity of the nodal gradient vectors via a multibody dynamic theory. The reason why we chose ANCF-ICE for the proposed framework is that the formulation has orthonormal vectors as nodal variables. The orthonormal vectors can be directly used to obtain the propeller cylinder coordinate system. On the other hand, Teixeira and Cesnik [1] used a strain-based beam formulation. If this formulation is used, the vectors need to be reconstructed from the strain variables. A 24-DoF ANCF beam element [25,28–30] is used in this paper because of the slender configuration of the wing. This element has two nodes A and B, as shown in Fig. 5. The generalized nodal coordinate vector of this element is defined as

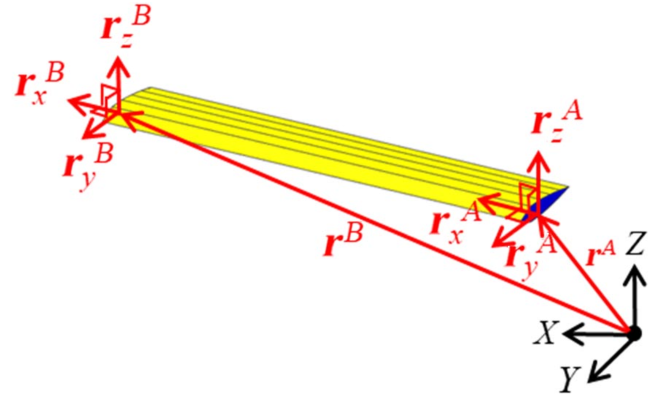


Fig. 5 ANCF-ICE beam element with orthogonal and unit gradient vectors.

$$\boldsymbol{e} \equiv \left[(\boldsymbol{r}^A)^T \;\; (\boldsymbol{r}_x^A)^T \;\; (\boldsymbol{r}_y^A)^T \;\; (\boldsymbol{r}_z^A)^T \;\; (\boldsymbol{r}^B)^T \;\; (\boldsymbol{r}_x^B)^T \;\; (\boldsymbol{r}_y^B)^T \;\; (\boldsymbol{r}_z^B)^T \right]^T \tag{7}$$

Here $\boldsymbol{r}$ is a position vector of a node, and $\boldsymbol{r}_x$, $\boldsymbol{r}_y$, and $\boldsymbol{r}_z$ are the gradient vectors to express the direction of the element. The use of the gradient vectors instead of rotation parameters provides a constant mass matrix and simple description of joints at the expense of a larger system of equations. Although the equation for one element is presented in this section for simplicity, a matrix superposition enables us to apply the equation to an arbitrary number of bodies with an arbitrary number of elements.

The norm of the gradient vectors compulsorily describes the cross-sectional extension that is negligible in the motion of high-aspect-ratio wings. Additionally, the cross-sectional deformation causes locking problems and the difficulty in applying the ANCF beam element to wings. To solve the problems, the ICEs to constrain the cross-sectional deformation were imposed in the previous study [30]. The resultant equation of ANCF-ICE is a differential algebraic equation (DAE) defined as

$$\boldsymbol{M}\ddot{\boldsymbol{e}} + \boldsymbol{F}_{\text{elastic}} + \left(\frac{\partial}{\partial \boldsymbol{e}} \begin{bmatrix} \boldsymbol{\Phi}_{\text{ICE}} \\ \boldsymbol{\Phi}_{\text{ECE}} \end{bmatrix} \right)^T \begin{bmatrix} \boldsymbol{\lambda}_{\text{ICE}} \\ \boldsymbol{\lambda}_{\text{ECE}} \end{bmatrix} = \boldsymbol{F}, \quad \begin{bmatrix} \boldsymbol{\Phi}_{\text{ICE}} \\ \boldsymbol{\Phi}_{\text{ECE}} \end{bmatrix} = \boldsymbol{0} \tag{8}$$

Here $\boldsymbol{M}$ is a constant mass matrix. $\boldsymbol{\Phi}_{\text{ICE}}$ and $\boldsymbol{\lambda}_{\text{ICE}}$ are the constraint vector and Lagrange's multiplier, respectively, related to the ICEs that enforce constraining the negligible cross-sectional deformations. Equally, $\boldsymbol{\Phi}_{\text{ECE}}$ and $\boldsymbol{\lambda}_{\text{ECE}}$ are the constraint vector and Lagrange's multiplier related to the ECEs to express mechanical joints. Because of the ICEs, the gradient vectors are orthogonal and unit, and both properties are leveraged to generate a propeller cylinder coordinate system efficiently, as will be shown in Sec. IV.A. To perform the dynamic analysis, the multibody dynamic theory to differentiate the constraint equation with respect to time yields

$$\begin{bmatrix} \boldsymbol{M} & \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ICE}}}{\partial \boldsymbol{e}}\right)^T & \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ECE}}}{\partial \boldsymbol{e}}\right)^T \\ \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ICE}}}{\partial \boldsymbol{e}}\right) & \boldsymbol{0} & \boldsymbol{0} \\ \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ECE}}}{\partial \boldsymbol{e}}\right) & \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \ddot{\boldsymbol{e}} \\ \boldsymbol{\lambda}_{\text{ICE}} \\ \boldsymbol{\lambda}_{\text{ECE}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{F} - \boldsymbol{F}_{\text{elastic}} \\ \boldsymbol{\gamma}_{\text{ICE}} \\ \boldsymbol{\gamma}_{\text{ECE}} \end{bmatrix} \tag{9}$$

The ECEs are linear functions of generalized nodal coordinates when they represent typical joints (e.g., fixed, hinge, and ball joints).

To perform the nonlinear static analysis using a Newton–Raphson method, the linearization of Eq. (8) without the inertial term is necessary. The external force $\boldsymbol{F}$ is assumed to be constant in a Newton–Raphson iteration. Therefore, only elastic force needs to be linearized around the given displacement $\boldsymbol{e}_0$ as

$$[\boldsymbol{K}]_{\boldsymbol{e}=\boldsymbol{e}_0} \equiv \left[\frac{\partial \boldsymbol{F}_{\text{elastic}}}{\partial \boldsymbol{e}}\right]_{\boldsymbol{e}=\boldsymbol{e}_0} \tag{10}$$

The equation used for the Newton–Raphson iteration in the nonlinear static analysis is

$$\begin{bmatrix} \boldsymbol{K} & \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ICE}}}{\partial \boldsymbol{e}}\right)^T & \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ECE}}}{\partial \boldsymbol{e}}\right)^T \\ \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ICE}}}{\partial \boldsymbol{e}}\right) & \boldsymbol{0} & \boldsymbol{0} \\ \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ECE}}}{\partial \boldsymbol{e}}\right) & \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}_{\boldsymbol{e}=\boldsymbol{e}_0} \begin{bmatrix} \Delta \boldsymbol{e} \\ \Delta \boldsymbol{\lambda}_{\text{ICE}} \\ \Delta \boldsymbol{\lambda}_{\text{ECE}} \end{bmatrix} = -\begin{bmatrix} [\boldsymbol{F}_{\text{elastic}}]_{\boldsymbol{e}=\boldsymbol{e}_0} + \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ICE}}}{\partial \boldsymbol{e}}\right)^T \boldsymbol{\lambda}_{\text{ICE}} + \left(\dfrac{\partial \boldsymbol{\Phi}_{\text{ECE}}}{\partial \boldsymbol{e}}\right)^T \boldsymbol{\lambda}_{\text{ECE}} - \boldsymbol{F} \\ [\boldsymbol{\Phi}_{\text{ICE}}]_{\boldsymbol{e}=\boldsymbol{e}_0} \\ [\boldsymbol{\Phi}_{\text{ECE}}]_{\boldsymbol{e}=\boldsymbol{e}_0} \end{bmatrix} \tag{11}$$

where the right-hand side is the residual force.

IV. Interaction of Propellers in Aeroelastic Model

In the nonlinear static and dynamic analyses considering propellers, the position and orientation of the propeller cylinder coordinate system can drastically change due to the large deformation of the wing. This section presents the effective method to derive the propeller cylinder coordinate system from the variables of the structural model.

A. Propeller Cylinder Coordinate Generation

A finite element node with a local coordinate system consisting of the gradient vectors $\boldsymbol{r}_x^{P0}$, $\boldsymbol{r}_y^{P0}$, and $\boldsymbol{r}_z^{P0}$ is selected to generate the propeller cylinder coordinate system, as shown in Fig. 6. To determine the propeller cylinder coordinate system, three parameters are introduced. The first parameter is the position vector of the propeller center defined as $\boldsymbol{r}_L^{P1} = \begin{bmatrix} 0 & y_L^{P1} & z_L^{P1} \end{bmatrix}$ in the local coordinate system. The second parameter is the pitch angle α of a propeller axis around $\boldsymbol{r}_x^{P1}$. The third parameter is the yaw angle β of a propeller axis around $\boldsymbol{r}_z^{P2}$. The propeller center is assumed to be located on the $\boldsymbol{r}_y^{P0}$–$\boldsymbol{r}_z^{P0}$ plane. The position vector of the propeller center is transformed in the global coordinate description as

$$\boldsymbol{r}^{P1} = \boldsymbol{r}^{P0} + \boldsymbol{J}_1 \boldsymbol{r}_L^{P1} \tag{12}$$

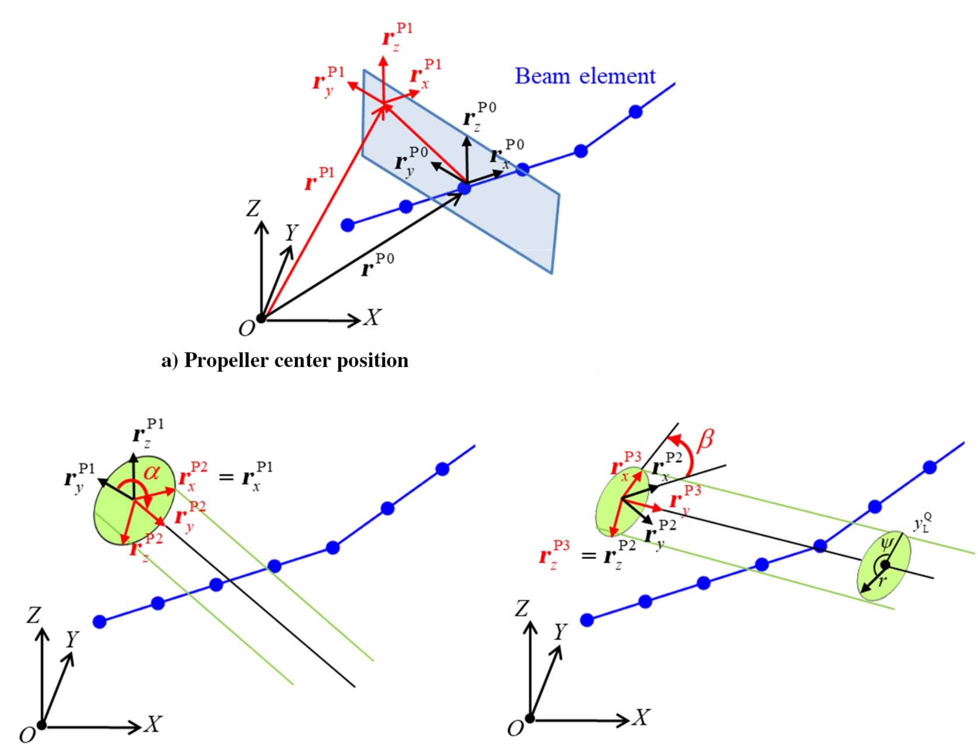
Here, the ICEs lead to the orthogonality and unity of the gradient vectors. Therefore, the transformation matrix from the local to global coordinate systems can be obtained directly from the nodal gradient vectors without using an orthonormalization process as

$$\boldsymbol{J}_1 \equiv \begin{bmatrix} \boldsymbol{r}_x^{P0} & \boldsymbol{r}_y^{P0} & \boldsymbol{r}_z^{P0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{r}_x^{P1} & \boldsymbol{r}_y^{P1} & \boldsymbol{r}_z^{P1} \end{bmatrix} \tag{13}$$

The base vectors $\boldsymbol{r}_i^{P2} (i = x, y, \text{and } z)$ of the local coordinate system considering the pitch angle α are obtained by rotating the local coordinate system around $\boldsymbol{r}_x^{P1}$. Then, the base vectors $\boldsymbol{r}_i^{P3} (i = x, y, \text{and } z)$ of the local coordinate system considering the yaw angle β are obtained by rotating the local coordinate system around $\boldsymbol{r}_z^{P2}$. Because the propeller cylinder axis is parallel to $\boldsymbol{r}_y^{P3}$, the flow velocity normal to the propeller plane is calculated as

$$U = (\boldsymbol{r}_y^{P3})^T (\boldsymbol{U} - \dot{\boldsymbol{r}}^{P1}) \tag{14}$$

Here, the one-way coupling assumption enables us to neglect the velocities induced by the bound and wake panels. Inputting this normal velocity into the BEM solver yields the induction factor that results in the propeller cylinder circulations. Here, the propeller-induced velocity at an arbitrary point Q with a position vector $\boldsymbol{r}^Q$ is considered. The position vector is defined in the propeller local coordinate system as



a) Propeller center position

b) Pitch

c) Yaw

Fig. 6 Cylinder coordinate generation.

$$\boldsymbol{r}_L^Q \equiv \begin{bmatrix} x_L^Q \\ y_L^Q \\ z_L^Q \end{bmatrix} = \boldsymbol{J}_3^T(\boldsymbol{r}^Q - \boldsymbol{r}^{P1}) \tag{15}$$

where the transformation matrix is also easily obtained without using the orthonormalization process as

$$\boldsymbol{J}_3 \equiv \begin{bmatrix} \boldsymbol{r}_x^{P3} & \boldsymbol{r}_y^{P3} & \boldsymbol{r}_z^{P3} \end{bmatrix} \tag{16}$$

The propeller cylinder coordinates y, r, and ψ can be defined as

$$\begin{aligned} y &\equiv y_L^Q, \\ r &\equiv \sqrt{(x_L^Q)^2 + (z_L^Q)^2}, \\ \psi &\equiv \begin{cases} \cos^{-1}(x_L^Q/r) & (z \geq 0) \\ 2\pi - \cos^{-1}(x_L^Q/r) & (z < 0) \end{cases} \end{aligned} \tag{17}$$

Substituting these cylinder coordinates and the propeller cylinder circulations into Eq. (1) yields the propeller-induced velocities at the point Q defined in the cylinder coordinate system. The propeller-induced velocities need to be transformed to the global coordinate system to agree with the UVLM formulation used for the wing aerodynamics. The propeller-induced velocity defined in the global coordinate system is written as

$$\boldsymbol{w}^P \equiv \boldsymbol{J}_3(u_a \boldsymbol{e}_a + u_t \boldsymbol{e}_t) \tag{18}$$

where the base vectors are defined in the propeller local coordinate system.

B. Nonlinear Static and Dynamic Aeroelastic Analysis Considering the Propeller

Figure 7a shows the flowchart of the nonlinear static analysis considering the propeller. The nonlinear static analysis requires the time updating to consider wake shedding of wings. The aerodynamic solver calculates the aerodynamic force corresponding to a constant displacement $\boldsymbol{e}^n$ until the aerodynamic force converges to a specific value (i.e., $|\boldsymbol{F}_{\text{aero}}^{t+\Delta t} - \boldsymbol{F}_{\text{aero}}^{t}|/|\boldsymbol{F}_{\text{aero}}^{t+\Delta t}| < \varepsilon_1$). By using Eq. (11) and the converged aerodynamic force $\boldsymbol{F}_{\text{aero}}^{t+\Delta t}$, the Newton–Raphson method seeks for the converged displacement $\boldsymbol{e}^{n+1}$ with the residual less than a user-defined small value. With the converged displacement $\boldsymbol{e}^{n+1}$, this process is repeated until $|\boldsymbol{e}^{n+1} - \boldsymbol{e}^n|/|\boldsymbol{e}^{n+1}| < \varepsilon_2$ is satisfied. Figure 7b shows the flowchart of the nonlinear dynamic analysis considering the propeller. Both the explicit (e.g., Runge–Kutta method) and implicit (e.g., generalized-α method) integration methods can be used. Our implementation has the explicit Runge–Kutta method and the implicit generalized-α method. Although our implementation also has weak and strong coupling solvers for the implicit integration, the results that will be presented in the following sections were obtained from the weak coupling solver because the two solvers did not show a discrepancy. In both the static and dynamic solvers, the propeller-induced loads and flows are time-averaged values along one propeller rotation period because we employ BEM to calculate the propeller circulation. We consider that the time-averaged propeller effects on the aeroelastic response are suitable for the aircraft of our interest because the propeller rotates at a high speed, whereas the aircraft response is relatively slow. In other words, the flowfield of our interest has different time scales. One is a propeller rotational time scale. The other is an incoming flow (or flight motion) time scale. The variation in the former time scale does not have a large effect on that in the latter time scale. Therefore, the propeller-induced loads and flows are approximated as time-averaged values.

V. Aerodynamic Analysis of Propeller–Wing System

The proposed aerodynamic model coupling BEM/AVCM and UVLM is analyzed in this section. First, the proposed BEM/AVCM is compared with the more complex model [10] in an aerodynamic analysis using an isolated propeller with a simple geometry for validation purposes. Then, the propeller is changed to an actual propeller for which experimental characteristics are available and the steady aerodynamics of the propeller–wing system are studied. Finally, the proposed method is compared with a conventional vortex-based method applying UVLM to both the propeller and wing aerodynamics in an unsteady propeller–wing interaction analysis.

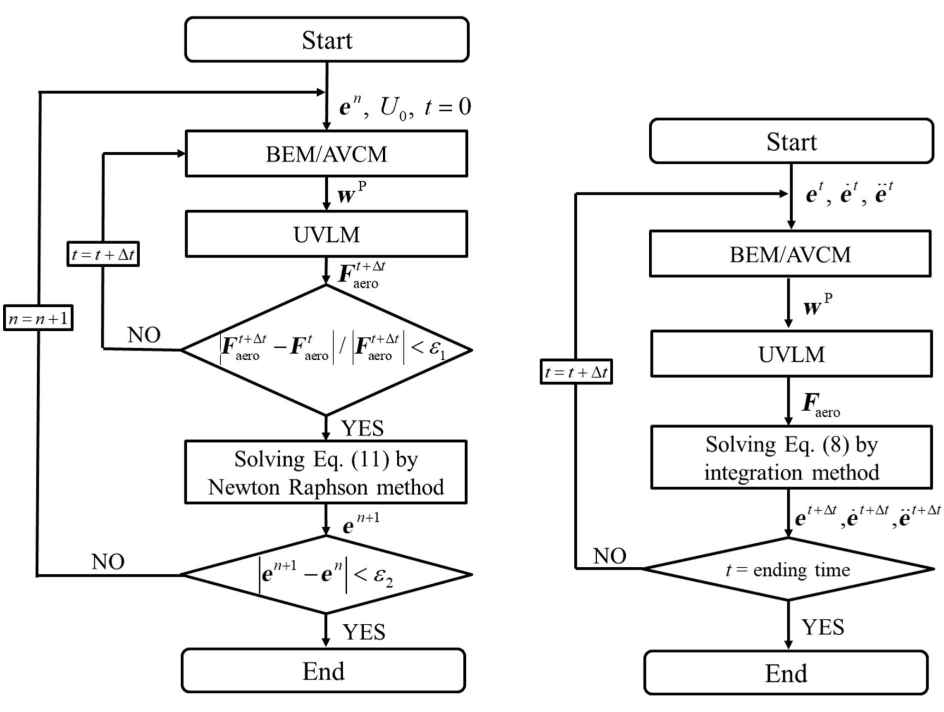


Fig. 7 Flowchart of nonlinear static and dynamic analyses considering propeller.

Table 1 Parameters of propeller with simple geometry used for verification analysis

Parameter	Value
Airfoil type	Flat plate
Propeller radius R	0.15 m
Propeller chord	0.01 m
Propeller hub radius	0.02 m
Angle of blade with plane of rotation	10 deg
Propeller rotation speed	6000 rpm
Number of blades	1

A. Verification of BEM/AVCM Using Isolated Simple Propeller

The aerodynamic analysis of an isolated propeller with a simple geometry presented by Teixeira and Cesnik [10] is replicated by using the proposed BEM/AVCM. Table 1 lists the analysis condition of the propeller. This propeller consists of one blade with a flat plate, a constant chord, and a constant twist. It also has an offset that represents a propeller hub radius. We also performed this analysis by using UVLM for the propeller aerodynamics. Figure 8 shows the relationship between the thrust coefficient and the freestream velocity. The results of the proposed BEM/AVCM agree well with those of UVLM and the reference values using VVPM [10,38]. Figure 9 shows the circulation along the propeller blade with a freestream velocity of 5, 7.5, and 10 m/s, respectively. The results of the BEM/AVCM solver agrees with those of UVLM.

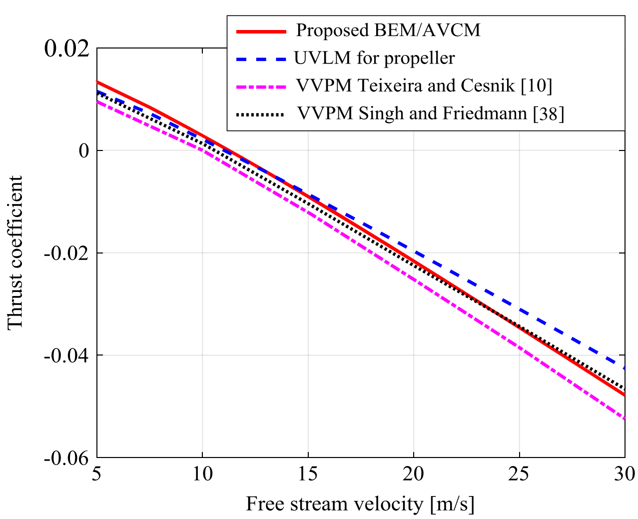


Fig. 8 Relationship between thrust coefficient and freestream velocity.

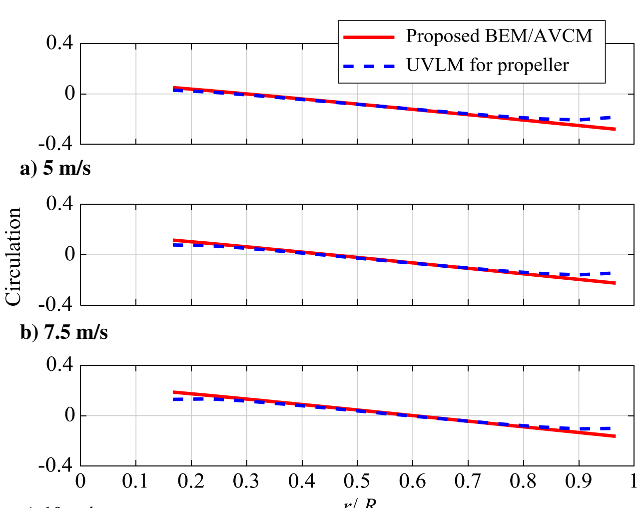


Fig. 9 Circulation distribution along propeller blade at a freestream velocity of a) 5 m/s, b) 7.5 m/s, and c) 10 m/s.

A large part of the computation is used for the propeller aerodynamics because the low flight speed and high propeller rotation speed require many propeller wake elements when vortex lattice, panel, and particle methods are used. On the other hand, the proposed method does not depend on the wake discretization. The calculation times are compared between the proposed BEM/AVCM and a conventional wake updating approach UVLM used for the propeller aerodynamics. The calculation time ratio used for one propeller rotation is calculated as

$$\text{Calculation time ratio} \equiv \frac{\text{Calculation time}^{\text{UVLM}}}{\text{Calculation time}^{\text{BEM/AVCM}}} \tag{19}$$

Figure 10 shows the relationship between the ratio and the number of the wake elements used for the UVLM. The ratio increases as the number of the wake elements increases. When aircraft has a large number of propellers, the calculation time ratio would become much larger.

To investigate the effect of the distance between the propeller and the wing (or the wing chord length), we compared the proposed BEM/AVCM and the conventional vortex-based method applying UVLM to the propeller with respect to the relationship between the propeller axial coordinate y and the mean flow velocity $\bar{U}$ on the vortex cylinder cross section. The uniform flow velocity is set to 5 m/s. The relative error of the proposed and conventional vortex-based methods is computed, and it is below 6%. The error does not change largely even if y/R increases. Therefore, the proposed method can be used for various distances between the propeller and the wing (or various wing chord lengths).

B. Verification of Propeller–Wing Steady Aerodynamic Interaction

The aerodynamics of an actual propeller is used for the propeller–wing interaction. The propeller is an Advanced Precision Composite (APC) thin electric propeller actually manufactured and used in experiments [39,40]. Table 2 lists the propeller conditions used in this analysis. The conditions were also used in the Refs. [1,9]. The aerodynamic coefficients of the propeller airfoil NACA4412 necessary for the BEM method were taken from [39] and approximated as

$$C_l = 0.10\alpha + 0.315, \quad C_d = 0 \tag{20}$$

The absolute value of the drag coefficient is much smaller than that of the lift coefficient especially at a low angle of attack, as shown in [39]. Therefore, we approximated the drag coefficient as zero. The propeller geometry information, namely, chord and twist distributions along the blade, necessary for the BEM method was

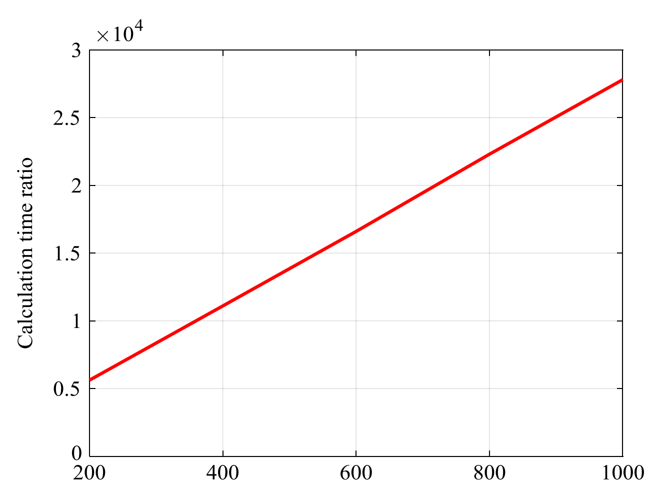


Fig. 10 Calculation time ratio of the proposed and conventional vortex-based methods in a propeller rotation of the one-blade propeller.

Table 2 Parameters of propeller and wing used for propeller–wing steady aerodynamic analysis

Parameter of propeller	Value of propeller
Airfoil type	NACA 4412
Propeller radius R	0.14 m
Propeller geometry (chord and twist distribution)	APC $11 \times 5.5E$ taken from [40]
Propeller rotation speed	6000 rpm
Number of blades	2
Air density	$1.225\ \mathrm{kg/m^3}$
Freestream velocity	14 m/s
Parameter of X-HALE rigid wing	**Value of X-HALE rigid wing**
Airfoil type	EMX-07
Wing semispan L	3 m
Wing chord	0.2 m
Position of wing axis (from leading edge)	29% chord
Dihedral wing position	$X = 2$ m
Dihedral wing angle	10 deg
Propeller–wing ratio R/L	0.047

taken from [40]. The propeller blade is discretized in 17 elements, as performed in [1,9]. The thrust and thrust coefficient calculated by the proposed BEM/AVCM are $T = 1.72$ N and $C_T = T/(\rho_a \times (6000/60)^2 \times (2R)^4) = 0.023$. These values agree well with $T = 1.68$ N and $C_T = 0.022$ calculated by the LL/VVPM in [1,9]. The small discrepancy is attributed to the difference between the aerodynamic coefficients used for the methods. The analysis object of the propeller–wing interaction is the main wing of a representative high-aspect-ratio-wing aircraft called X-HALE [1,9]. As done in Ref. [1], the wing is treated as a rigid wing in this section to focus on the steady aerodynamics of the propeller–wing interaction. Table 2 also lists the parameters of the wing. The wing has dihedral members, as shown in Fig. 11a. The analysis conditions are the same as those used in Sec. V.B in [1]. The midpoint of the wing span is fixed to the origin of the global coordinate system with an angle of attack 2 deg plus an incidence angle 5 deg. The wing has five APC propellers at 1 m intervals in the span direction, as shown in Fig. 11a. The propellers are located 0.2 m ahead and 0.028 m below of the leading edge of the wing. The propeller pitch angle is $\alpha = 175$ deg. The two propellers attached to the right wing rotate in the clockwise direction. The other three propellers rotate in the opposite direction. The propeller–wing ratio R/L is 0.047. To compare the proposed BEM/AVCM propeller model with the LL/VVPM propeller model [1,9], the wing is discretized in the same UVLM panels used in [1,9] (i.e., 8 chordwise and 36 spanwise panels for the semispan). Figures 11b and 11c show the lift coefficients along the wing axis without and with propellers, respectively. The results of the proposed method are compared with those of the Refs. [1,9]. In the comparative analysis without the propeller (i.e., comparison of UVLM codes) in Fig. 11b, the results of our analysis and the reference are found to be in very good agreement although there is a small discrepancy of the lift coefficient distribution along the dihedral members. This small discrepancy is probably attributed to the difference between the calculation methods for the local lift coefficient considering the dihedral angle. Figure 11c shows the lift coefficient distribution considering the propellers. The accelerated flow due to the propeller-induced axial velocity increases the total lift. The upwash and downwash due to the propeller-induced tangential velocity change the spanwise lift distribution, but they do not change the total lift. The proposed method provides a slightly larger lift coefficient along the wing span behind the propellers than the reference because the propeller circulation that is proportional to the propeller thrust calculated by BEM/AVCM is slightly larger than that of LL/VVPM, as mentioned above. The reasonable agreement of the lift coefficient distribution demonstrates the applicability of the proposed BEM/AVCM averaging the circulation and propeller-induced tangential velocity under the condition of $R/L = 0.047$. The propeller–wing ratio $R/L = 0.027$ of Helios Prototype HP03 is smaller than

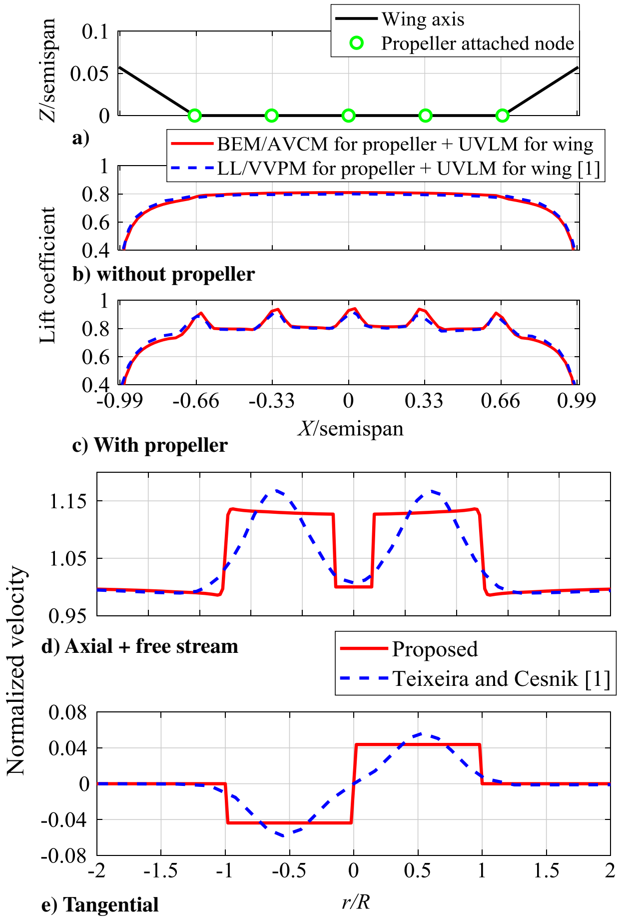


Fig. 11 a) Wing axis of X-HALE main wing, b) lift coefficients without propellers, c) lift coefficient with propellers, d) propeller-induced axial velocity profile, and e) propeller-induced tangential velocity profile.

$R/L = 0.047$. Therefore, the proposed BEM/AVCM is applicable to the aircraft of our interest.

Figures 11d and 11e show the velocity profiles behind a propeller. Although the constant propeller circulation assumption of the proposed method provides averaged velocity profiles, the assumption has a smaller effect on the wing aerodynamics when the propeller–wing ratio R/L is small because the lift distributions on the main wing are in good agreement between the proposed method and the method of Teixeira and Cesnik [1], as shown in Fig. 11c.

C. Verification of Propeller–Wing Unsteady Aerodynamic Interaction

We compare the proposed method with the conventional vortex-based method applying a UVLM to both the propeller and wing aerodynamics in an unsteady analysis. To focus more on the unsteady effect of the propeller–wing interaction, we use a propeller–wing system with a simpler geometry. Table 3 shows the parameters of the propeller–wing system. Five propellers are attached to the wing at 1 m intervals in the span direction. The propellers are located 0.2 m ahead of the leading edge of the wing. The propeller–wing system undergoes a heaving motion. Before the comparative verification, convergence tests were performed with respect to the number of spanwise panels N for the semispan, the number of chordwise panels M, and the time step dt. A root mean square (RMS) error of the lift coefficient is calculated as

$$\mathrm{Error} \equiv \frac{\mathrm{RMS}(C_L - C_L^{\text{fine discretization}})}{\max(C_L)} \times 100[\%] \tag{21}$$

Here $dt = 0.001$, $N = 72$, and $M = 16$ are used as fine discretization values that are two times finer than those used in the same wing

Table 3 Parameters of propeller and wing used for propeller–wing unsteady aerodynamic analysis

Parameters of propeller in Table 1 are used.	
Parameter of a simple rigid wing	Value of a simple rigid wing
Airfoil type	Flat
Wing semispan L	3 m
Wing chord	0.2 m
Position of wing axis (from leading edge)	50% chord
Dihedral wing position	No dihedral
Dihedral wing angle	0 deg
Angle of attack	0 deg
Heaving frequency	0.6 Hz
Heaving amplitude	0.3 m
Propeller–wing ratio R/L	0.05

size [1]. Figure 12 shows convergence test results. Figure 12a shows a fluctuated result with respect to N. When the propeller diameter is smaller than the spanwise panel size, most collocation points on the wing are not located inside the vortex cylinder. In such a case, the propeller-induced flow on the wing and the maximum C_L are underestimated. For example, the maximum C_L at $N = 12$ is 89.4% of the maximum C_L at $N = 72$. On the other hand, if N is a moderately high value, a collocation point is inside the cylinder, whereas nearly half of the panel is outside of the cylinder, as shown in Fig. 4. In such a case, the lift is overestimated. For example, the maximum C_L at $N = 21$ is 103% of the maximum C_L at $N = 72$. As described in Sec. II.B, this overestimation is mitigated by two methods. The first method is to increase N. Here, the maximum C_L at $N = 36$ is 96% of the maximum C_L at $N = 72$. The second method is to evaluate the propeller-induced flow at four corner points of a vortex panel, and then the mean value of the velocities is defined as the propeller-induced velocity at a collocation point. However, the mitigation effect of the second method is less than 1%.

According to the convergence tests, $N = 48$ and $M = 14$ are used for the comparison between the proposed and conventional vortex-based methods. Although $dt = 0.01$ s provides a well converged solution of the proposed method, the conventional wake-updating method requires a small dt to express the high-speed-rotating wake of the propellers. Therefore, $dt = 0.002$ s is employed for the comparison.

Figure 13 shows the time histories of the lift coefficient. The proposed and conventional vortex-based methods are in good agreement. Because the propeller-induced axial velocity increases the local lift, the total lift coefficient increases when the propellers are attached to the wing. Here, to remove the singularity effect of UVLM caused by the collision of the rotating propeller wake panels and the wing panels, a low-pass filter was applied to the conventional UVLM result. Although the conventional vortex-based method using UVLM for the propeller can express the propeller wake deformation caused by a quick sideslip or strong gusts, it requires careful treatment for the singularity and a high computational cost. Therefore, the proposed method that has been validated in the propeller–wing steady and unsteady aerodynamic analyses is more useful from the viewpoint of its computational efficiency and easy singularity avoidance.

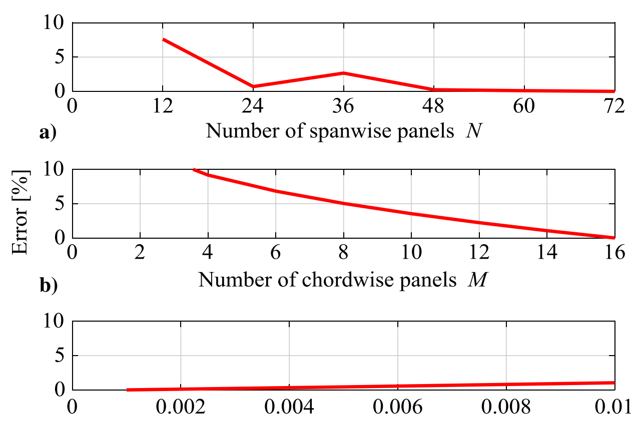


Fig. 12 Convergence test results for the a) number of spanwise panels N, b) number of chordwise panels M, and c) time step.

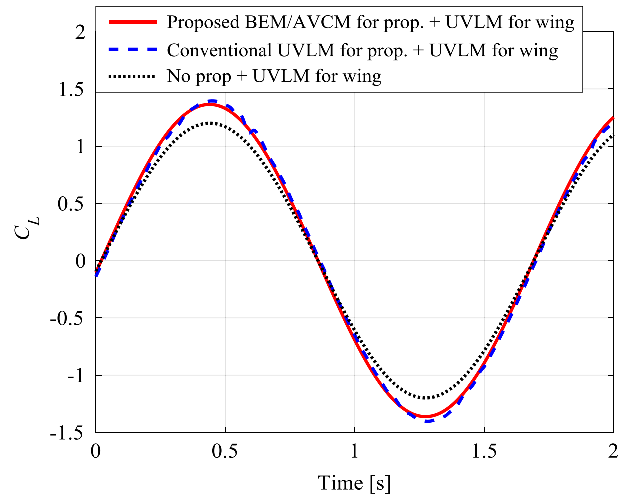


Fig. 13 Time histories of the lift coefficients in unsteady heaving motion.

VI. Aeroelastic Analysis of Propeller–Wing System

This section presents nonlinear static and dynamic analyses using the coupled propeller–wing aeroelastic model proposed in Sec. IV.

To the best of the authors' knowledge, the only reference data for the nonlinear aeroelastic analysis with propellers can be found in the work of Teixeira and Cesnik [1]. However, direct comparison with the reference is difficult due to the complexity of the complete aircraft model used in [1] and the differences in terms of code implementations.

Although we do not perform the direct comparison in the aeroelastic analyses with the propellers, nonlinear aeroelastic analyses without the propellers were validated by conducting the wind tunnel experiment in the author's previous study [31]. In addition, the proposed method was found to be in good agreement with the conventional vortex-based methods in the steady and unsteady propeller–wing aerodynamic analyses in Sec. V. These separated comparisons insist that the proposed method is valid in the nonlinear aeroelastic analysis with the propellers. Furthermore, we will present results of the aeroelastic analysis with the propellers in Sec. VI.B that can be explained by the Theodorsen's unsteady aerodynamic theory. This also facilitates showing the reliability of the proposed method in the nonlinear aeroelastic analysis with the propellers.

In the nonlinear aeroelastic analyses, a high-aspect-ratio wing with a simpler geometry is used to focus on investigating the propeller–wing interactive effects on the nonlinear static and dynamic responses. This wing has been widely used as a benchmark test model [18,19,21,22,25,37]. Ten propellers are attached to the benchmark wing by assuming the configuration of the aircraft of our interest, such as Helios Prototypes. Table 4 lists the wing and propeller parameters used in this analysis. The propellers are located at $X = -14, -12, -10, -8, -6, 6, 8, 10, 12$, and 14 m in the span direction of the undeformed state. The propellers are located 1 m ahead of the leading edge with a pitch angle of 180 deg. The vertical offset of the propellers with respect to the leading edge is zero. The propeller–wing ratio $R/L = 0.044$ used in this analysis is less than $R/L = 0.047$ used in the previous validation analysis. Thus, the proposed BEM/AVCM is applicable to this configuration with $R/L = 0.044$. Based on a convergence study, the semiwing is discretized in 32 structural elements. The wing chord is discretized in six panels, as performed in [19]. Although the semispan of the benchmark wing without propellers was discretized in 20 panels in [19], it is discretized in 64 panels in the current analysis to capture the propeller effects well. Even when the propellers were

Table 4 Parameters of propeller and benchmark wing used for propeller–wing aeroelastic analysis

Parameter of propeller	Value of propeller
Airfoil type	NACA 4412
Propeller radius R	0.7 m
Propeller geometry (chord and twist distribution)	APC $11 \times 5.5E$ taken from [40]
Propeller rotation speed	2143 rpm
Number of blades	2
Air density	$0.0889\ \mathrm{kg/m^3}$
Freestream velocity	25 m/s
Parameter of benchmark flexible wing	**Value of benchmark flexible wing**
Airfoil type	NACA 0012
Wing semispan L	16 m
Wing chord	1 m
Center of gravity (form leading edge)	50% chord
Elastic axis (form leading edge)	50% chord
Mass per unit length	0.75 kg/m
Moment of inertia (around elastic axis)	0.1 kg m
Extensional stiffness	$1 \times 10^7\ \mathrm{N \cdot m}$
Torsional stiffness	$2 \times 10^4\ \mathrm{N \cdot m^2}$
Spanwise bending stiffness	$1 \times 10^4\ \mathrm{N \cdot m^2}$
Chordwise bending stiffness	$4 \times 10^6\ \mathrm{N \cdot m^2}$
Propeller–wing ratio R/L	0.044

considered, the 64-spanwise-panel discretization provided a less than 1% error of the lift coefficient compared with a 128-spanwise-panel discretization.

A. Aeroelastic Static Analysis

The benchmark wing is cantilevered with a root angle of attack of 4 deg. The static analysis is performed according to Fig. 7a. Figure 14 shows a three-dimensional view of the deformed and undeformed states of the wing. The proposed method to generate the cylinder coordinate system enables the propeller planes to follow the large static deformation. Figure 15 shows the static deflections with and without the propellers. In the case without the propellers, the results of the proposed framework agree very well with the reference values obtained from a corotational beam formulation coupled with a UVLM [22]. The propeller increases the static deflection because the lift is increased, as shown in the previous section. The proposed method can split the propeller-induced velocity in axial and tangential components. The effects of each propeller-induced velocity are investigated. The deflection considering only axial velocity is 5.4% larger than that without the propellers. The local static lift is a quadratic function of the flow velocity increased by the propeller-induced axial velocity. On the other hand, the deflection considering only tangential velocity is 0.96% smaller than that without the propellers. The upwash and downwash tangential effects caused by adjacent propellers are canceled, and thus the noncanceled downwash effect of the most outer propeller decreases the deflection. The lift is a linear function of angle of attack decreased by the downwash effect. Therefore, the variation of the static deflection caused by the propeller-induced tangential velocity is smaller than that caused by the propeller-induced axial velocity.

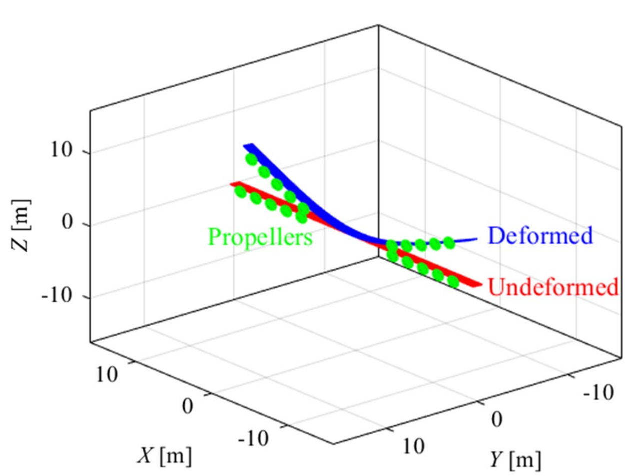


Fig. 14 Three-dimensional view of static deformation of wing with propellers.

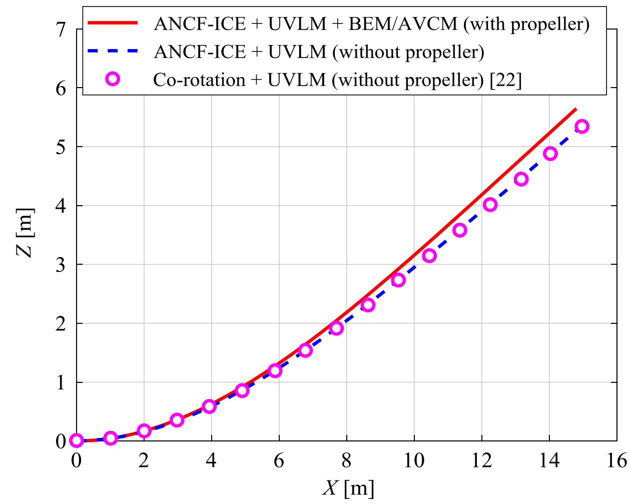


Fig. 15 Static deflections with and without propellers.

B. Aeroelastic Dynamic Analysis

The dynamic response of the wing with and without the propellers is compared for bending excitation. The benchmark wing is cantilevered with a root angle of attack of 4 deg at the flight condition described above. From the nonlinear static deformed state, a vertical force $100 \cdot \sin \omega t$ N applied to the wingtip excites the dynamic motion; ω is set to the natural angular frequency 2.24 rad/s of the first bending mode. Figure 16 shows the time histories of the wingtip bending with and without the propellers. The displacement is shifted in the positive direction by the propellers mainly because the static deflection is increased by the propeller-induced axial velocity. In Fig. 16, however, the discrepancy of the local maxima *MAX* is smaller than that of the local minima *MIN*. This is because the geometrical nonlinear stiffness increases as the displacement increases.

To explain the propeller effect on the dynamics further, the root angle of attack is reset to 0 deg to remove the static deflection, and the propeller-induced velocities are considered separately. Figure 17a

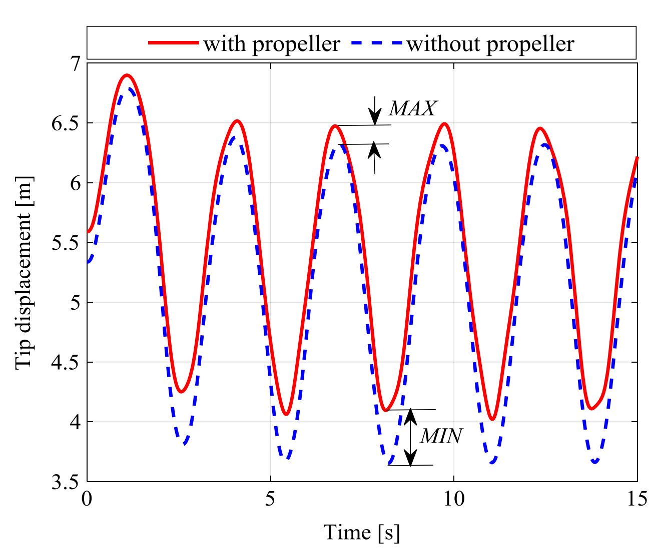


Fig. 16 Time history of dynamic wingtip displacement from the nonlinear static equilibrium.

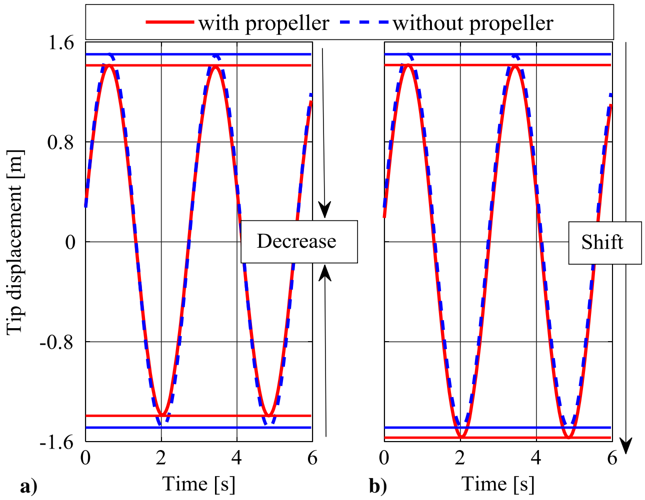


Fig. 17 Time history of tip displacement considering propeller-induced a) axial and b) tangential velocities.

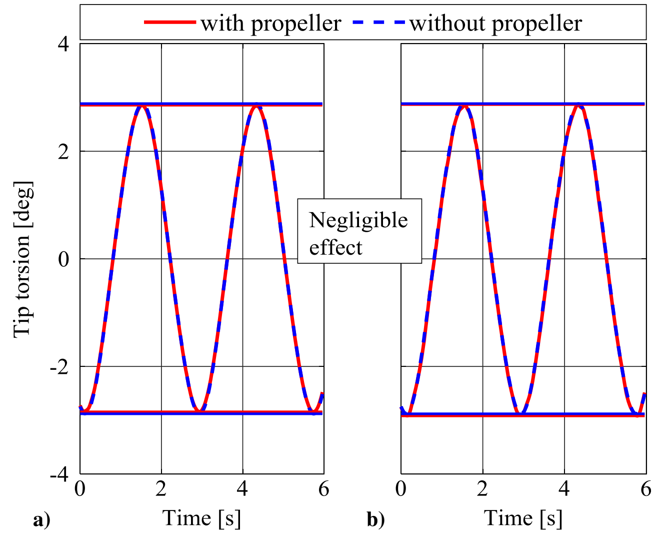


Fig. 18 Time history of tip torsion considering propeller-induced a) axial and b) tangential velocities.

shows the time histories of the tip bending with and without the propeller-induced axial velocity. The amplitude obtained from the analysis considering the propeller-induced axial velocity is 6.7% smaller than that without the propellers. To explain this damping mechanism, we use a classical Theodorsen's unsteady aerodynamic force with a quasi-steady assumption (i.e., Theodorsen's function is approximated as one) [41] because of its theoretical simplicity and relatively low-frequency of the high-aspect-ratio wing. We consider a rigid airfoil with two degrees of freedom, namely, heaving (i.e., bending) h and torsion α, supported by heaving and torsion springs. When this airfoil is subject to the Theodorsen's unsteady aerodynamic force with a quasi-steady assumption, the equation of motion can be written as

$$\begin{aligned}\boldsymbol{M}_{\text{airfoil}}\begin{bmatrix}\ddot{h}\\ \ddot{\alpha}\end{bmatrix}+\boldsymbol{K}_{\text{airfoil}}\begin{bmatrix}h\\ \alpha\end{bmatrix}&=\begin{bmatrix}A_{11}&A_{12}\\ A_{21}&A_{22}\end{bmatrix}\begin{bmatrix}\ddot{h}\\ \ddot{\alpha}\end{bmatrix}\\&+\begin{bmatrix}B_{11}&B_{12}\\ B_{21}&B_{22}\end{bmatrix}\begin{bmatrix}\dot{h}\\ \dot{\alpha}\end{bmatrix}+\begin{bmatrix}C_{11}&C_{12}\\ C_{21}&C_{22}\end{bmatrix}\begin{bmatrix}h\\ \alpha\end{bmatrix},\\&=\begin{bmatrix}-\pi\rho_a b^2&0\\ 0&-\frac{1}{8}\pi\rho_a b^4\end{bmatrix}\begin{bmatrix}\ddot{h}\\ \ddot{\alpha}\end{bmatrix}\\&+\begin{bmatrix}-2\pi\rho_a Ub&2\pi\rho_a Ub^2\\ -\pi\rho_a Ub^2&0\end{bmatrix}\begin{bmatrix}\dot{h}\\ \dot{\alpha}\end{bmatrix}+\begin{bmatrix}0&2\pi\rho_a U^2 b\\ 0&\pi\rho_a U^2 b^2\end{bmatrix}\begin{bmatrix}h\\ \alpha\end{bmatrix}\end{aligned} \tag{22}$$

Here, b is a half-chord. The flow velocity U in the term B_{11} increased by the propeller-induced axial velocity increases the aerodynamic damping, and thus the bending amplitude is decreased by the propellers. Figure 17b shows the time histories of the tip bending with and without the propeller-induced tangential velocity. The vibration center shifts in the negative direction when the propeller-induced tangential velocity is considered because the downwash of the most outer propeller causes a negative flow-induced angle of attack α, as shown in the previous static analysis. Although there is no structural bending–torsion coupling in this wing (i.e., the center of gravity is located on the elastic axis), the torsion is caused by the bending excitation because of the coupling term B_{21}, as shown in Fig. 18. Figure 18a shows that the damping effect of the propeller-induced axial velocity on the dynamic torsion is negligibly smaller than that on the dynamic bending in Fig. 17a. This is because the main damping term of torsion B_{22} is zero. Figure 18b shows that the effect of the propeller-induced tangential velocity on the dynamic torsion is negligibly small. This is because the coupling term C_{21} is zero although the downwash of the most outer propeller decreases the static bending [i.e., h in Eq. (22)].

VII. Conclusions

A nonlinear static and dynamic analysis framework for high-aspect-ratio wings with a computationally efficient model for the propellers has been described. In the aerodynamic modeling, the propeller wake is modeled as a vortex cylinder by taking advantage of the low-frequency dynamics of the high-aspect-ratio wings. By leveraging the small propeller–wing ratio R/L, the AVCM is proposed that enables a computationally efficient calculation of the propeller-induced velocities and avoids the numerical singularities without the loss of accuracy once the propeller blade circulation is provided. The propeller blade circulation was calculated by the BEM theory. The results of the aerodynamic propeller–wing interaction analysis using the proposed BEM/AVCM were in very good agreement with the reference values obtained using the more complex methods.

An efficient cylinder coordinate generation method has also been proposed by leveraging the orthogonality and unity of the gradient vectors of the ANCF with the internal constraint equations (ANCF-ICE). Once the propeller-attached node is selected, the propeller cylinder coordinate system to calculate the propeller-induced velocities is obtained from the nodal gradient vectors in a straightforward and systematic manner even when the wing undergoes large deformation.

The proposed method can also split the propeller-induced velocity in axial and tangential components, which has successfully demonstrated that the propeller-induced axial velocity has a larger effect on the variation in the static deflection than the tangential velocity because the lift is a quadratic function of the axial velocity, whereas it is a linear function of the tangential velocity (i.e., flow-induced angle of attack).

By using the proposed method and a classical unsteady aerodynamic theory, the propeller effects on the dynamic responses have been explained. The dynamic analysis demonstrated that the propeller damps the bending amplitude rather than the torsion amplitude in the bending excitation. This is because the main damping term of the bending is increased by the propeller-induced axial velocity, whereas that of the torsion is initially zero. The effect of the propeller-induced tangential velocity on the dynamic bending is negligibly small because the bending changed by the tangential velocity does not cause a coupling with torsion.

The application range of the proposed method with respect to some factors was not presented explicitly in this paper because the comparative data do not exist in this new research field and the focus of this paper is the proposal of a new analysis framework. The expected cases where the proposed method would not be accurate are close propellers, strong gust, and quick sideslip. As future studies in this aeroelastic research field considering the propeller, it is necessary to investigate the application range of the proposed method by building a higher-fidelity simulation model or conducting an experiment.

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